

Link Model Deployment AWS: <https://youtu.be/7MKUf5lloqc>

Link Model Deployment Lokal: <https://youtu.be/XZcUd3loBQw>

Link Github AWS: <https://github.com/NysaSetiawan/projekakhirmd>

Perbandingan

Apabila lokal, maka biaya tidak ada tidak seperti AWS yang menggunakan dolar dan dihitung berdasarkan penggunaan

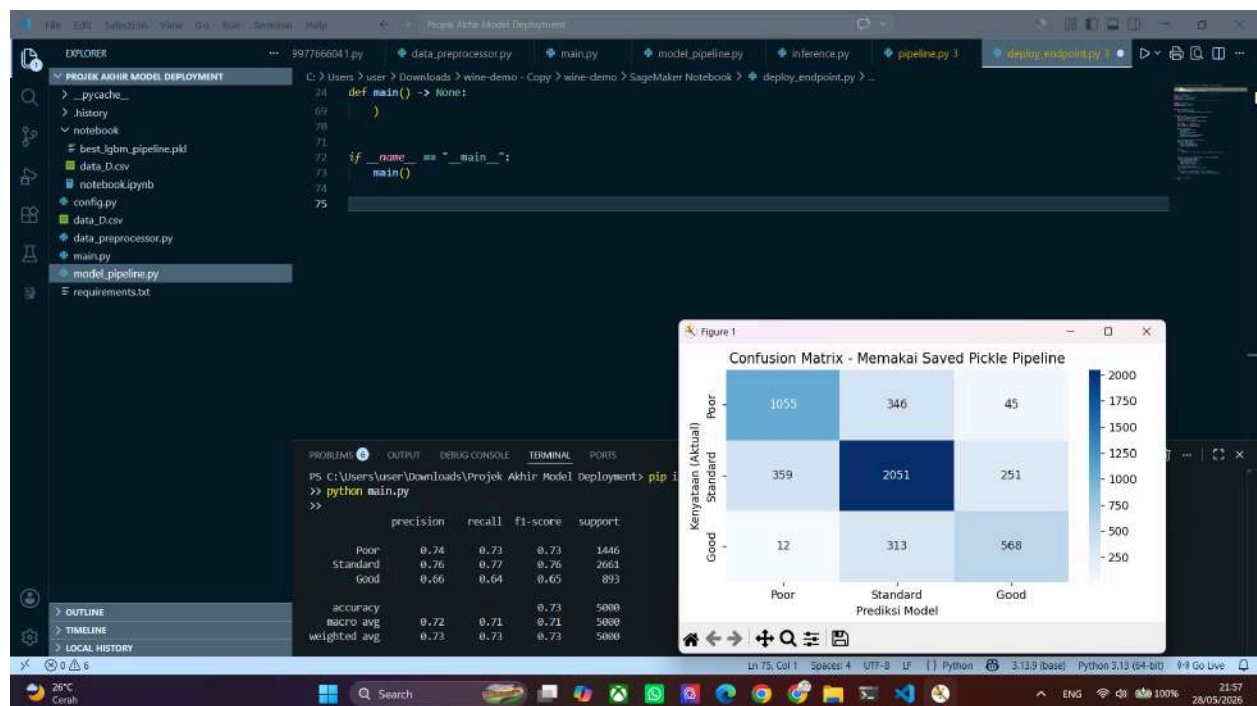
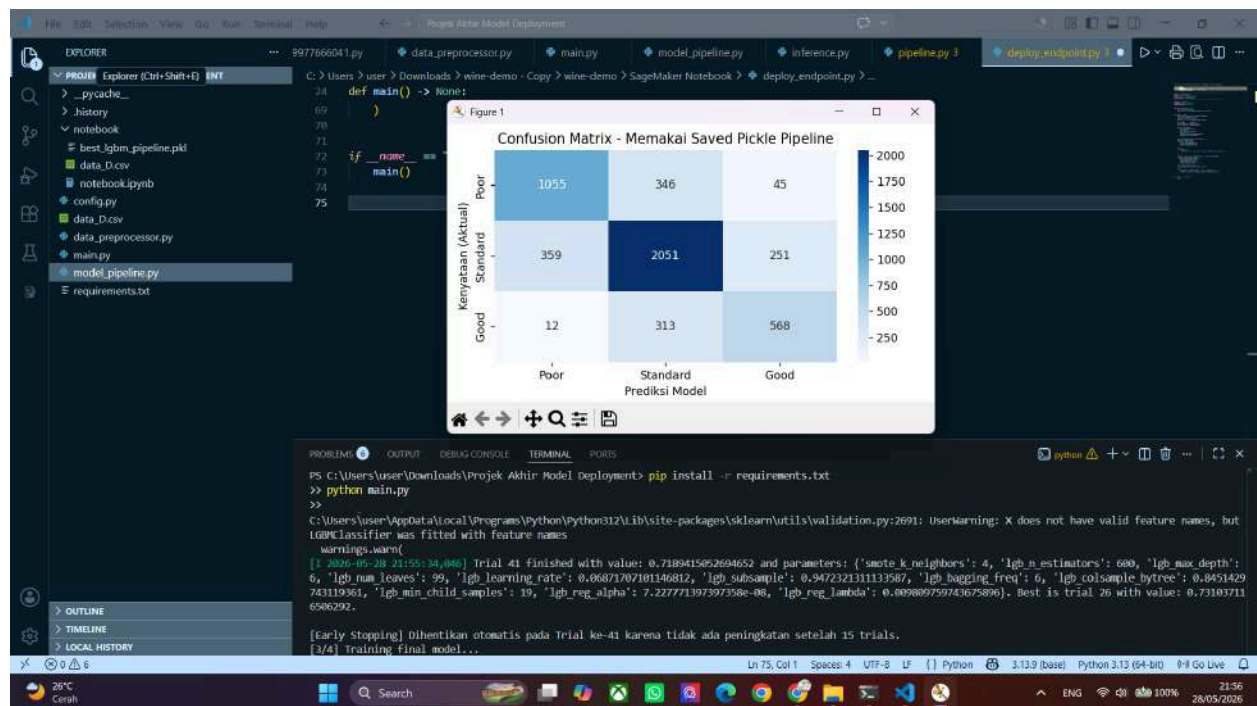
Apabila lokal, hanya bisa diakses menggunakan komputer atau jaringan lokal sedangkan cloud bisa diakses secara online dengan endpoint API

Apabila lokal, keamanannya bergantung pada pengaturan kita, sedangkan di cloud dapat kita atur di IAM Role nya

Apabila lokal, bergantung pada laptop sendiri untuk misalnya CPU, GPU sedangkan cloud disediakan dan dapat kita pilih instance nya

Apabila lokal, error harus dicari sendiri, sedangkan di cloud dapat dimonitor melalui cloudwatch

Lokal



`pip install -r requirements.txt`

`python main.py`

`streamlit run streamlit.py`

Command prompt

```
C:\Users\user\anaconda3\Scripts\activate base
Mlflow ui
```

Di terminal, jalankan kode ini. Dari sini, dapat terlihat bahwa pipeline saya yakni [main.py](#) berhasil dijalankan dan berhasil mengeluarkan hasil beserta confusion matrix.

Streamlit Lokal: Standard

Credit Approval Prediction

Please fill out the customer metrics below to evaluate their creditworthiness instantly.

Personal Information	Financial Information	Loan & Behavioral Profile
Customer ID CUS_0x697f	Number of Bank Accounts 2	Changed Credit Limit (\$) 10,00
ID 0x45c7	Number of Active Credit Cards 3	Number of Credit Inquiries 1
Customer Name Nysa Setiawan	Credit Interest Rate (%) 12,00	Credit Mix Bad
SSN (Social Security Number) 312-56-4208	Number of Active Loans 1	Outstanding Debt (\$) 1500,00
Evaluation Month January	Types of Loans (Separate with commas or 'and') Personal Loan, Home Loan	Credit Utilization Ratio (%) 30,00

Credit Approval Prediction | Dokumentasi Projek Akhir Mo... | Ask Gemini

localhost:8501

Deploy

Evaluation Month: January

Types of Loans (Separate with commas or "and"): Personal Loan, Home Loan

Credit Utilization Ratio (%): 30,00

Age: 30

Average Delay from Due Date (Days): 5

Credit History Age (Text format): 22 Years and 4 Months

Occupation: Scientist

Number of Delayed Payments: 2

Paying Minimum Amount Only?: No

Annual Income (\$): 50000,00

Monthly EMI Assessment (\$): 300,00

Monthly in-hand Salary (\$): 4000,00

Amount Invested Monthly (\$): 100,00

Payment Behaviour: Low_spent_Small_value_payments

Monthly Balance (\$): 500,00

Calculate Customer Credit Score

Credit Approval Prediction | Dokumentasi Projek Akhir Mo... | Ask Gemini

localhost:8501

Deploy

Low_spent_Small_value_payments

Monthly Balance (\$): 500,00

Calculate Customer Credit Score

Prediction Evaluation Summary:

Customer Credit Score Rank: STANDARD (Moderate Risk Category)

Classification Probabilities:

	Poor	Standard	Good
0	15.03%	81.56%	3.41%

Bad

32 WhatsApp

Menjalankan Aplikasi

NysaSetiawan/projek

Dokumentasi Projek A

Credit Approval Predi

Quick Play — Play-CS

Ask Gemini

localhost:8501

Deploy

Credit Approval Prediction

Please fill out the customer metrics below to evaluate their creditworthiness instantly.

Personal Information	Financial Information	Loan & Behavioral Profile
Customer ID CUS_0x697f	Number of Bank Accounts 2	Changed Credit Limit (\$) 10,00
ID 0x45c7	Number of Active Credit Cards 3	Number of Credit Inquiries 5
Customer Name Nysa Setiawan	Credit Interest Rate (%) 30,00	Credit Mix Bad
SSN (Social Security Number) 312-56-4208	Number of Active Loans 1	Outstanding Debt (\$) 222200,00
Evaluation Month January	Types of Loans (Separate with commas or 'and') Personal Loan, Home Loan	Credit Utilization Ratio (%) 30,00
Age	Average Delay from Due Date (Days)	Credit History Age (Text format)

localhost:8501

Deploy

Age 30	Average Delay from Due Date (Days) 5	Credit History Age (Text format) 22 Years and 4 Months
Occupation Scientist	Number of Delayed Payments 10	Paying Minimum Amount Only? No
Annual Income (\$) 100,00		Monthly EMI Assessment (\$) 1,00
Monthly In-hand Salary (\$) 20,00		Amount Invested Monthly (\$) 10,00
		Payment Behaviour Low_spent_Small_value_payments
		Monthly Balance (\$) 500,00

Calculate Customer Credit Score

localhost:8501

Deploy

Low_spent_Small_value_payments

Monthly Balance (\$)

500,00

Calculate Customer Credit Score

Prediction Evaluation Summary:

Customer Credit Score Rank: POOR (High Default Risk Category)

Classification Probabilities:

	Poor	Standard	Good
0	68.10%	31.27%	0.63%

Good

Credit Approval Prediction

Please fill out the customer metrics below to evaluate their creditworthiness instantly.

Personal Information

Customer ID

CUS_0x697f

ID

0x45c7

Customer Name

Nysa Setiawan

SSN (Social Security Number)

312-56-4208

Evaluation Month

January

Financial Information

Number of Bank Accounts

3

Number of Active Credit Cards

3

Credit Interest Rate (%)

1,00

Number of Active Loans

1

Types of Loans (Separate with commas or 'and')

Personal Loan, Home Loan

Loan & Behavioral Profile

Changed Credit Limit (\$)

10,00

Number of Credit Inquiries

5

Credit Mix

Good

Outstanding Debt (\$)

200,00

Credit Utilization Ratio (%)

10,00

Credit Approval Prediction

localhost:8501

Deploy

Age

30

-

+

Average Delay from Due Date (Days)

1

-

+

Credit History Age (Text Format)

22 Years and 4 Months

Occupation

Engineer

Number of Delayed Payments

1

-

+

Paying Minimum Amount Only?

No

Annual Income (\$)

20000000,00

-

+

Monthly EMI Assessment (\$)

1,00

-

+

Monthly In-hand Salary (\$)

12222222,00

-

+

Amount Invested Monthly (\$)

20000,00

-

+

Payment Behaviour

Low_spent_Small_value_payments

Monthly Balance (\$)

11000,00

-

+

Calculate Customer Credit Score

Prediction Evaluation Summary:

28°C Berawan

Search

ENG 97%

20:32 31/05/2026

Prediction Evaluation Summary:

Customer Credit Score Rank: GOOD (High Trust Category)

Classification Probabilities:

	Poor	Standard	Good
0	0.49%	29.32%	70.19%

28°C Berawan

Search

ENG 97%

20:32 31/05/2026

Cloud SageMakerAI

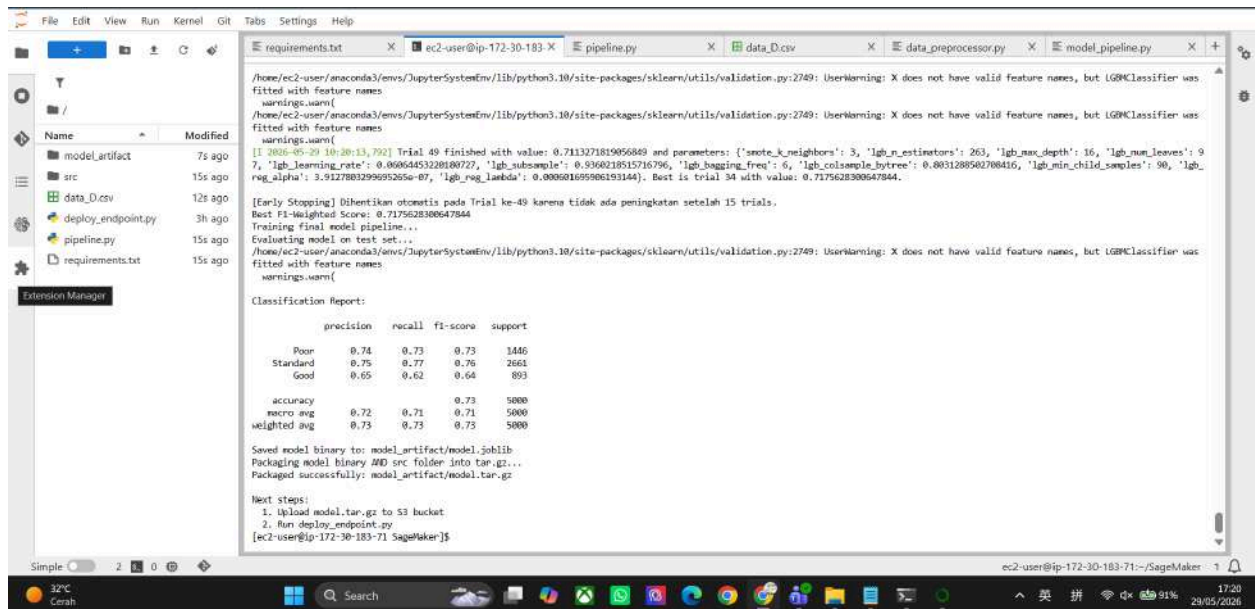
Cd SageMaker

Is

python pipeline.py

aws s3 cp model_artifact/model.tar.gz s3://<nama-bucket-anda>/credit/model.tar.gz

python deploy_endpoint.py



```
[ec2-user@ip-172-30-183-71 SageMaker]$ BUCKET=credit-demo-333
```

```
[ec2-user@ip-172-30-183-71 SageMaker]$ aws s3 mb s3://$BUCKET --region us-east-1
```

```
make_bucket: credit-demo-333
```

```
[ec2-user@ip-172-30-183-71 SageMaker]$ aws s3 cp model_artifact/model.tar.gz
```

```
s3://credit-demo-333/model/model.tar.gz
```

```
upload: model_artifact/model.tar.gz to s3://credit-demo-333/model/model.tar.gz
```

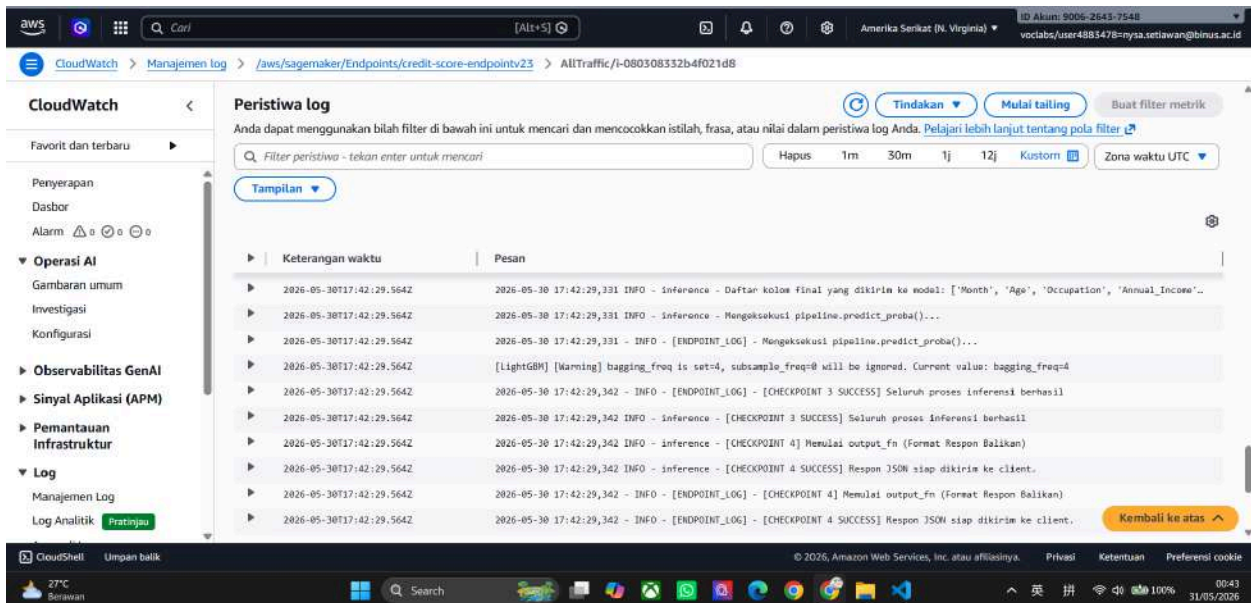
```
[ec2-user@ip-172-30-183-71 SageMaker]$
```

```
Successfully installed Django-2.2.1 certifi-2019.6.16 cycloper-0.10.0 kiwisolver-1.1.0 matplotlib-3.1.0 numpy-1.16.4 pypar
sing-2.4.0 python-dateutil-2.8.0 pytz-2019.1 six-1.12.0 sqlparse-0.3.0 tornado-6.0.3
WARNING: You are using pip version 19.2.3, however version 24.0 is available.
You should consider upgrading via the 'python -m pip install --upgrade pip' command.
```

```
(sagemaker-env) [ec2-user@ip-172-23-0-67 SageMaker]$ python deploy_endpoint.py
sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
Role ARN: arn:aws:iam::900626437548:role/LabRole
Model URI: s3://credit-demo-333/model/model.tar.gz

Deploying real-time endpoint...
-----!
Credit test response:
{"probabilities": [[0.0047372406537652244, 0.27073923107166836, 0.7245235282745665]], "predictions": [2], "labels": ["Good"]}
(sagemaker-env) [ec2-user@ip-172-23-0-67 SageMaker]$
```

Apabila ada error, dapat dilihat di cloudwatch. Saya rekomendasikan untuk membuat checkpoint untuk inference. Sehingga dapat melihat apakah ada kesalahan di checkpoint. Misal Checkpoint 3. Error.



```
(my_venv) [ec2-user@ip-172-21-137-201 SageMaker]$ deactivate
[ec2-user@ip-172-21-137-201 SageMaker]$ rm -rf /home/ec2-user/SageMaker/my_venv
[ec2-user@ip-172-21-137-201 SageMaker]$ sudo mkdir /my_venv_global
sudo chown ec2-user:ec2-user /my_venv_global
[ec2-user@ip-172-21-137-201 SageMaker]$ python3 -m venv /my_venv_global/my_venv
[ec2-user@ip-172-21-137-201 SageMaker]$ # Aktifkan venv dari lokasi baru
[ec2-user@ip-172-21-137-201 SageMaker]$ source /my_venv_global/my_venv/bin/activate
(my_venv) [ec2-user@ip-172-21-137-201 SageMaker]$ python -m pip install --no-cache-dir
--no-deps sagemaker lightgbm optuna imbalanced-learn pandas numpy scikit-learn==1.2.1
boto3 streamlit
```

```
source /my_venv_global/my_venv/bin/activate
python -m pip install --no-cache-dir --force-reinstall numpy==1.24.3 scikit-learn==1.2.1
```

```
(my_venv) [ec2-user@ip-172-21-137-201 SageMaker]$ pip show sagemaker
```

Name: sagemaker

Version: 2.230.0

Summary: Open source library for training and deploying models on Amazon SageMaker.

Home-page: <https://github.com/aws/sagemaker-python-sdk>

Author: Amazon Web Services

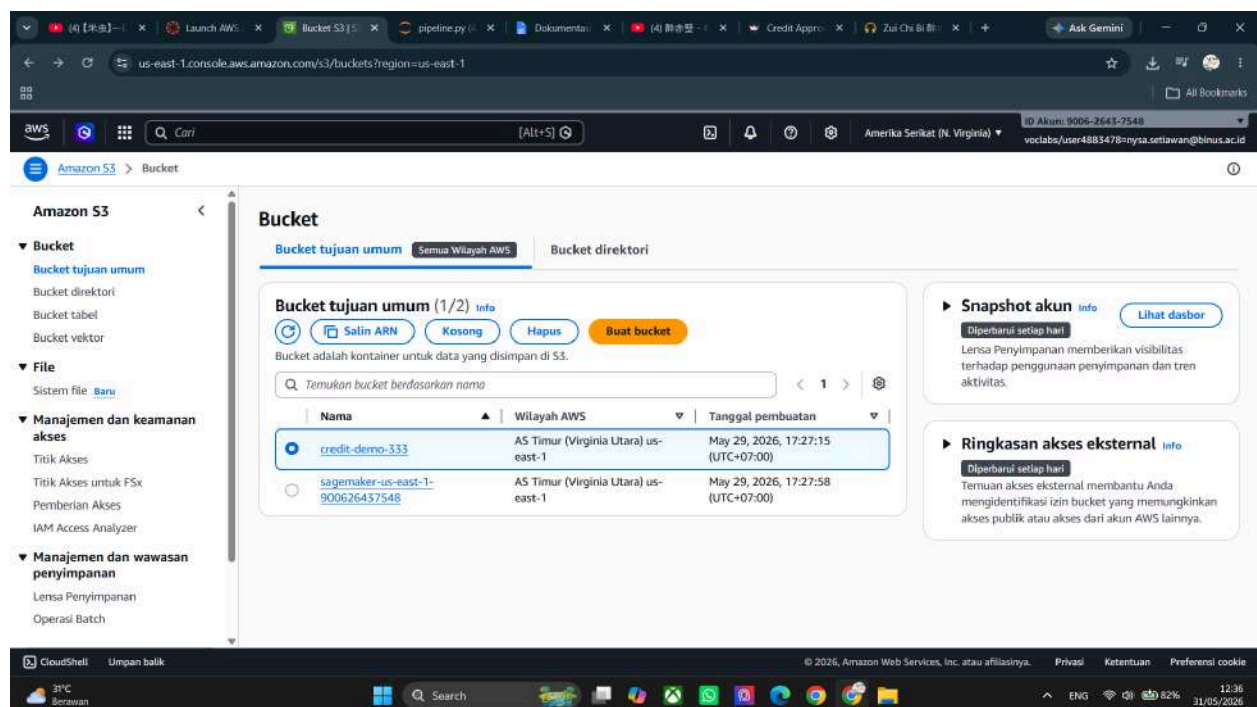
Author-email:

License:

Location: /my_venv_global/my_venv/lib/python3.10/site-packages

Requires: attrs, boto3, cloudpickle, docker, google-pasta, importlib-metadata, jsonschema, numpy, packaging, pandas, pathos, platformdirs, protobuf, psutil, pyyaml, requests, schema, smdebug-rulesconfig, tblib, tqdm, urllib3

Required-by:



The screenshot shows the Amazon S3 console interface. The left sidebar contains navigation options like 'Bucket', 'File', 'Manajemen dan keamanan akses', and 'Manajemen dan wawasan penyimpanan'. The main content area is titled 'Bucket' and shows a table of buckets. The table has columns for 'Nama', 'Wilayah AWS', and 'Tanggal pembuatan'. Two buckets are listed: 'credit-demo-333' and 'sagemaker-us-east-1-900626457548'. The 'sagemaker-us-east-1-900626457548' bucket is selected. To the right of the table, there are two informational boxes: 'Snapshot akun' and 'Ringkasan akses eksternal'.

Nama	Wilayah AWS	Tanggal pembuatan
credit-demo-333	AS Timur (Virginia Utara) us-east-1	May 29, 2026, 17:27:15 (UTC+07:00)
sagemaker-us-east-1-900626457548	AS Timur (Virginia Utara) us-east-1	May 29, 2026, 17:27:58 (UTC+07:00)

us-east-1.console.aws.amazon.com/sagemaker/home?region=us-east-1#/notebooks-and-git-repos

Amazon SageMaker AI > Notebook dan Repositori Git

AI Amazon SageMaker

Dashboard
What's new

▼ **Environment configuration**

Domain
Citra
Konfigurasi siklus hidup
Manajer peran

▼ **Applications and IDEs**

SageMaker Studio
Canvas
RStudio
Notebooks
TensorBoard
Profiler
Partner AI Apps

Luncurkan notebook dalam hitungan detik dan mulai pengodean secara instan
Gunakan komputasi dan penyimpanan dasar yang serupa dengan instans notebook AWS untuk mengaktifkan lebih banyak fitur dengan biaya yang sama
Jalankan alur kerja ML dan analitik yang komprehensif dengan lancar, semuanya dalam satu notebook
Manfaatkan bantuan pengodean yang didukung GenAI dari Amazon Q Developer dan JupyterAI untuk mempercepat pengembangan
Berkolaborasi dengan rekan Anda dalam waktu nyata di notebook yang sama untuk penemuan ide yang lancar

▶ Bagaimana cara mengakses JupyterLab di Studio?

Instans Notebook | Repositori Git

Instans notebook info

Temukan instans notebook

Nama	Instans	Waktu pembuatan	Status	Tindakan
☒ uasplissfinal	ml.m5.xlarge	31/5/2026, 08.07.26	✓ InService	Buka Jupyter Buka JupyterLab
☐ uasplastibisa	ml.m5.xlarge	30/5/2026, 19.14.14	⊖ Stopped	Mulai

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31°C
Berawan

12:16
31/05/2026

aws CloudWatch > Manajemen log > /aws/sagemaker/Endpoints/credit-score-endpoint27 > AllTraffic/i-0cd92aa8f4f3b1292

CloudWatch

Favorit dan terbaru

Penyerapan
Dasbor
Alarm

▼ **Operasi AI**

Gambaran umum
Investigasi
Konfigurasi

► **Observabilitas GenAI**

► **Sinyal Aplikasi (APM)**

► **Pemantauan Infrastruktur**

▼ **Log**

Manajemen Log
Log Analitik [Praktikum](#)

Peristiwa log

Anda dapat menggunakan bilah filter di bawah ini untuk mencari dan mencocokkan istilah, frasa, atau nilai dalam peristiwa log Anda. [Pelajari lebih lanjut tentang pola filter](#)

Filter peristiwa - tekan enter untuk mencari

Hapus 1m 30m 1j 12j Kustom Zona waktu UTC

Tampilkan

Keterangan waktu	Pesan
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,773 - INFO - [ENDPOINT_LOG] - Daftar kolom final yang dikirim ke model: ['Month', 'Age', 'Occupation', 'Annual_Income...]
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,773 - INFO - [ENDPOINT_LOG] - Mengeksekusi pipeline.predict_proba()...
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,773 INFO - Inference - Daftar kolom final yang dikirim ke model: ['Month', 'Age', 'Occupation', 'Annual_Income...]
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,773 INFO - Inference - Mengeksekusi pipeline.predict_proba()...
2026-05-31T03:23:04.979Z	[LightGBM] [Warning] bagging_freq is set=4, subsample_freq=0 will be ignored. Current value: bagging_freq=4
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,784 - INFO - [ENDPOINT_LOG] - [CHECKPOINT 3 SUCCESS] Seluruh proses inferensi berhasil
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,784 - INFO - [ENDPOINT_LOG] - [CHECKPOINT 4] Memulai output_fn (Format Respon Balikan)
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,784 INFO - Inference - [CHECKPOINT 3 SUCCESS] Seluruh proses inferensi berhasil
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,784 INFO - Inference - [CHECKPOINT 4] Memulai output_fn (Format Respon Balikan)
2026-05-31T03:23:04.979Z	2026-05-31 03:23:04,784 INFO - Inference - [CHECKPOINT 4 SUCCESS] Respon JSON siap dikirim ke client.

Kembali ke atas

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31°C
Berawan

10:24
31/05/2026

```
File Edit View Run Kernel Git Tabs Settings Help
+
Name Modified
model_artifact 1h ago
src 10m ago
+2.12.0 1h ago
data_D.csv 2h ago
deploy_endpoint.py 5m ago
error.log 33m ago
pipeline.py 10m ago
streamlit_app.py now
user-data.sh 2h ago

2. Run deploy_endpoint.py
(my_venv) [ec2-user@ip-172-21-137-201 SageMaker]$ aws s3 cp model_artifact/model.tar.gz s3://credit-demo-333/model/model.tar.gz
upload: model_artifact/model.tar.gz to s3://credit-demo-333/model/model.tar.gz
(my_venv) [ec2-user@ip-172-21-137-201 SageMaker]$ python deploy_endpoint.py
sagemaker.config INFO - Not applying SDK defaults from location: /etc/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
Role ARN: arn:aws:iam::900626437548:role/LabRole
Model URI: s3://credit-demo-333/model/model.tar.gz

Deploying real-time endpoint...
Traceback (most recent call last):
  File "/home/ec2-user/SageMaker/deploy_endpoint.py", line 67, in <module>
    main()
  File "/home/ec2-user/SageMaker/deploy_endpoint.py", line 48, in main
    predictor = model.deploy(
  File "/my_venv_global/my_venv/lib/python3.10/site-packages/sagemaker/model.py", line 1753, in deploy
    self.sagemaker_session.endpoint_from_production_variants(
  File "/my_venv_global/my_venv/lib/python3.10/site-packages/sagemaker/session.py", line 5818, in endpoint_from_production_variants
    self.sagemaker_client.create_endpoint_config(**config_options)
  File "/my_venv_global/my_venv/lib/python3.10/site-packages/botocore/client.py", line 606, in _api_call
    return self._make_api_call(operation_name, kwargs)
  File "/my_venv_global/my_venv/lib/python3.10/site-packages/botocore/context.py", line 123, in wrapper
    return func(*args, **kwargs)
  File "/my_venv_global/my_venv/lib/python3.10/site-packages/botocore/client.py", line 1094, in _make_api_call
    raise error_class(parsed_response, operation_name)
botocore.exceptions.ClientError: An error occurred (ValidationException) when calling the CreateEndpointConfig operation: Cannot create already existing endpoint configuration "arn:aws:sagemaker:us-east-1:900626437548:endpoint-config/credit-score-endpoint/26".
(my_venv) [ec2-user@ip-172-21-137-201 SageMaker]$ python deploy_endpoint.py
sagemaker.config INFO - Not applying SDK defaults from location: /etc/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
Role ARN: arn:aws:iam::900626437548:role/LabRole
Model URI: s3://credit-demo-333/model/model.tar.gz

Deploying real-time endpoint...
-----
Credit test response:
{"probabilities": [[0.00039738274544292867, 0.10976473304041363, 0.8898378842101434]], "predictions": [2], "labels": ["Good"]}
(my_venv) [ec2-user@ip-172-21-137-201 SageMaker]$
```

EC2

Instans

Dasbor

AWS Global View

Peristiwa

▼ Instans

Instans

Type Instans

Templat Peluncuran

Permintaan Spot

Savings Plans

Instans Terpesan

Host Khusus

Reservasi Kapasitas

Capacity Manager

▼ Citra

AMI

Katalog AMI

▼ Penyimpanan Blok

Elastic

Instans (1/4) info

Hubungkan

Status instans

Tindakan

Luncurkan instans

Set filter tersimpan

Pilih set filter

Temukan instans berdasarkan atribut atau tanda (case-sensitive)

	Name	ID Instans	Status instans	Tipe instans	Pemeriksaan stats	Availability Zone	DNS IPv4 publik	
☐	credit app nysa	i-07ef0427a6e5acefa	Menjalankan	t3.micro	3/3 pemeriksaan	us-east-1b	ec2-54-226-80-156.co...	5
☐	credit	i-05a9bee8342176591	Menjalankan	t3.micro	3/3 pemeriksaan	us-east-1b	ec2-54-165-33-98.com...	5
☐	Credit Pasti Fl...	i-062aed847369c59d8	Diakhiri	t3.micro	-	us-east-1b	-	-
☒	Credit Paling ...	i-0d12bbb94414440b3	Menjalankan	t3.micro	3/3 pemeriksaan	us-east-1b	ec2-34-204-44-251.co...	3

i-0d12bbb94414440b3 (Credit Paling Oke)

Rincian

Status dan alarm

Pemantauan

Keamanan

Jaringan

Penyimpanan

Tanda

▼ Ringkasan instans info

ID Instans

i-0d12bbb94414440b3

Alamat IPv4 publik

34.204.44.251 | buka alamat

Alamat IPv4 privat

172.31.18.85

CloudShell

Umpun balik

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Privasi

Ketentuan

Preferensi cookie

The image consists of two screenshots of the AWS Management Console, specifically the EC2 Instances page. The top screenshot shows a list of four EC2 instances, all with a status of 'disabled'. The bottom screenshot shows the detailed view for the instance with ID 'i-0d12bbb94414440b3', which is in the 'Running' state. The details are organized into several sections: General information, Network settings, and IAM/Storage settings.

Top Screenshot: EC2 Instances List

Pemantauan	Nama grup keamanan	Nama kunci	Waktu peluncuran	Rincian ...	Dikelola	Operator
disabled	launch-wizard-6	-	2026/05/31 11:01 GMT+7	Linux/UNIX	salah	-
disabled	launch-wizard-5	-	2026/05/31 11:01 GMT+7	Linux/UNIX	salah	-
disabled	-	-	2026/05/31 11:21 GMT+7	Linux/UNIX	salah	-
disabled	launch-wizard-8	-	2026/05/31 11:34 GMT+7	Linux/UNIX	salah	-

Bottom Screenshot: Instance Details for i-0d12bbb94414440b3

Ringkasan instans untuk i-0d12bbb94414440b3 (Credit Paling Oke)

Diperbarui less than a minute lalu

ID Instans
i-0d12bbb94414440b3

Alamat IPv6
-

Tipe hostname
Nama IP: ip-172-31-18-85.ec2.internal

Jawab nama DNS sumber daya privat
IPv4 (A)
34.204.44.251 [IP Publik]

Alamat IP yang ditetapkan secara otomatis
34.204.44.251 [IP Publik]

Peran IAM
LabRole

IMDSv2
Required

Alamat IPv4 publik
34.204.44.251 | buka alamat

Status instans
Menjalankan

Nama DNS IP privat (hanya IPv4)
ip-172-31-18-85.ec2.internal

Tipe instans
t3.micro

ID VPC
vpc-055278c154726f813

ID Subnet
subnet-02dd7743d02543e63

ARN Instans
arn:aws:ec2:us-east-1:900626437546:instance/i-0d12bbb94414440b3

Alamat IPv4 privat
172.31.18.85

DNS Publik
ec2-34-204-44-251.compute-1.amazonaws.com | buka alamat

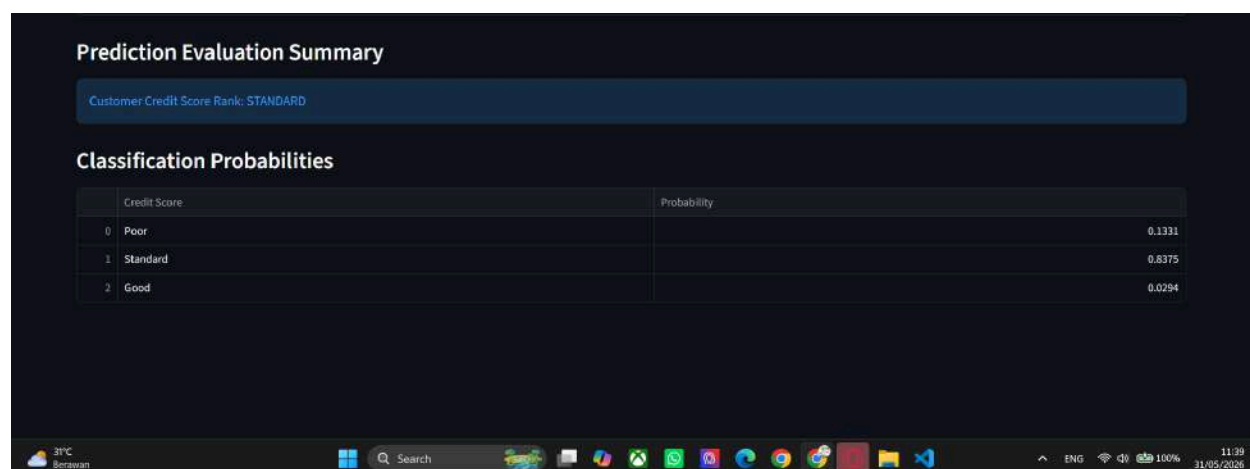
Alamat IP Elastis
-

Pencarian AWS Compute Optimizer
Pilih AWS Compute Optimizer untuk rekomendasi. | Pelajari selengkapnya

Nama Grup Auto Scaling
-

Dikelola
salah

Dokumentasi Keberhasilan:
Standard



POOR

Credit Approval Prediction

Personal Information	Financial Information	Loan & Behavioral Profile
Record Number (Unnamed: 0)	Number of Bank Accounts	Changed Credit Limit (\$)
1	2	10,00
ID	Number of Active Credit Cards	Number of Credit Inquiries
0x2b4a	4	10
Customer ID	Credit Interest Rate (%)	Credit Mix
CUST-9999	30,00	Bad
Customer Name	Number of Active Loans	Outstanding Debt (\$)
Nysa Setiawan	6	1500,00
SSN	Types of Loans	Credit Utilization Ratio (%)
000-00-0000	Personal Loan, Home Loan	30,00
Evaluation Month	Average Delay from Due Date (Days)	Credit History Age (Text Format)
January	10	22 Years and 4 Months

31°C Borawan 11:40 31/05/2026

Browser tabs: user-di, projek, Launch, Instans, Dokum, (4) 群, New To, Credit, Credit, Ask Gemini

Address bar: Not secure 34.204.44.251:8501

Calculate Customer Credit Score

Age: 55, Number of Delayed Payments: 2, Paying Minimum Amount Only?: No

Occupation: Unknown, Monthly EMI Assessment (\$): 300,00

Annual Income (\$): 100,00, Amount Invested Monthly (\$): 100,00

Monthly In-hand Salary (\$): 200,00, Payment Behaviour: High_spent_Large_value_payments

Monthly Balance (\$): 100,00

Browser tabs: user-di, projek, Launch, Instans, Dokum, (4) 群, New To, Credit, Credit, Ask Gemini

Address bar: Not secure 34.204.44.251:8501

Calculate Customer Credit Score

100,00

Prediction Evaluation Summary

Customer Credit Score Rank: POOR

Classification Probabilities

	Credit Score	Probability
0	Poor	0.9362
1	Standard	0.0622
2	Good	0.0017

Taskbar: 31°C Borawan, Search, ENG, 100%, 11:40 31/05/2026

Good

4 [R] x user-di x proje x Launch x Instans x Dokum x (4) 群 x New To x Credit x Credit x + Ask Gemini

← → ↻ Not secure 34.204.44.251:8501 ☆ 📄 🧑 All Bookmarks

Credit Approval Prediction

Personal Information

Record Number (Unnamed: 0)

1 - +

ID

0x2b4a

Customer ID

CUST-9999

Customer Name

Nysa Setiawan

SSN

000-00-0000

Evaluation Month

January

Financial Information

Number of Bank Accounts

2 - +

Number of Active Credit Cards

1 - +

Credit Interest Rate (%)

10,00 - +

Number of Active Loans

1 - +

Types of Loans

Personal Loan, Home Loan

Average Delay from Due Date (Days)

2 - +

Loan & Behavioral Profile

Changed Credit Limit (\$)

10,00 - +

Number of Credit Inquiries

2 - +

Credit Mix

Good

Outstanding Debt (\$)

100,00 - +

Credit Utilization Ratio (%)

10,00 - +

Credit History Age (Text format)

1 Years and 0 Months

Age

30 - +

Occupation

Engineer

Annual Income (\$)

122000,00 - +

Monthly In-hand Salary (\$)

7000,00 - +

Number of Delayed Payments

0 - +

Paying Minimum Amount Only?

No

Monthly EMI Assessment (\$)

300,00 - +

Amount Invested Monthly (\$)

100,00 - +

Payment Behaviour

High_spent_Large_value_payments

Monthly Balance (\$)

100,00 - +

Calculate Customer Credit Score

Prediction Evaluation Summary

Customer Credit Score Rank: GOOD

31°C Berawan

Search

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ENG 📶 🔋 100%

11:43 31/05/2026

← → ↻ ⚠ Not secure 34.204.44.251:8501

☆ 📄 👤 ⋮

🔍 All Bookmarks

100,00 - +

Calculate Customer Credit Score

Prediction Evaluation Summary

Customer Credit Score Rank: GOOD

Classification Probabilities

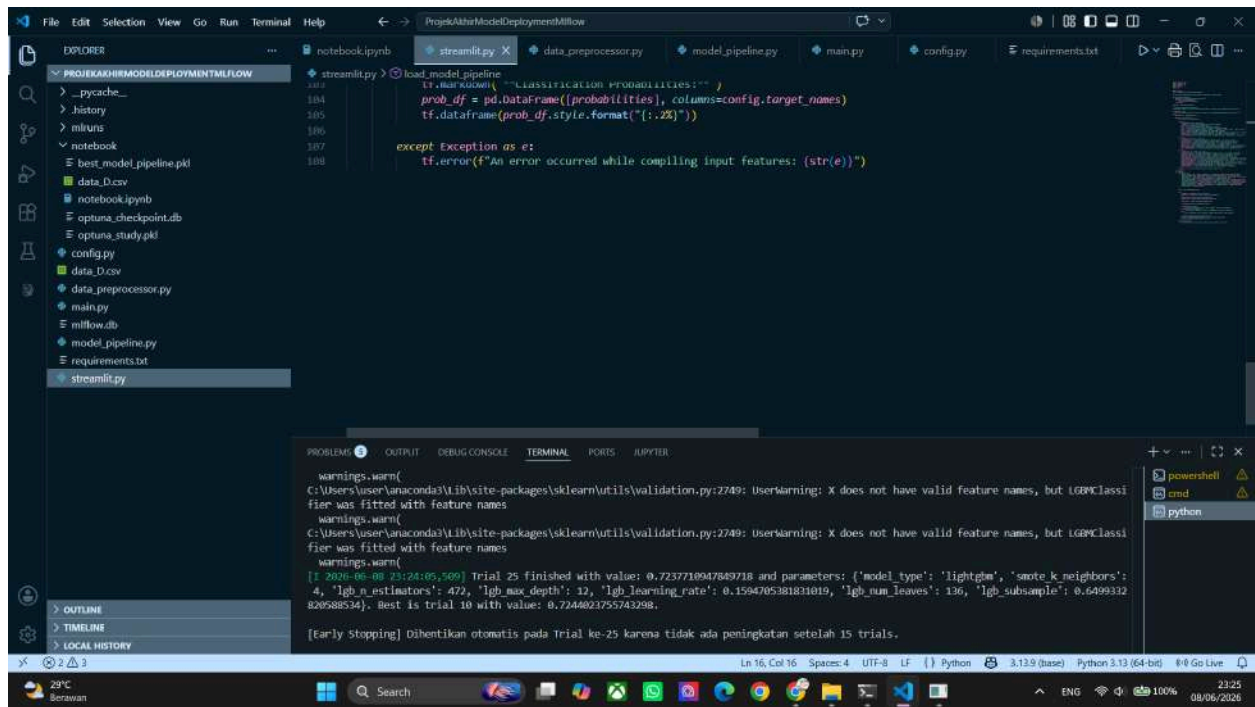
	Credit Score	Probability
0	Poor	0.00003
1	Standard	0.0198
2	Good	0.9801

31°C
Beraikan

ENG 📶 🔋 100%

11:44
31/05/2026

Lokal Part 2 bagian .py dan mlflow ui:



```
File Edit Selection View Go Run Terminal Help
ProjectAlphaModelDeploymentMlflow
EXPLORER
PROJEKTAHModelDeploymentMlflow
  .__pycache__
  .history
  .mlruns
  notebook
    best_model_pipeline.pkl
    data_D.csv
    notebook.ipynb
    optuna_checkpoint.db
    optuna_study.pkl
  config.py
  data_D.csv
  data_preprocessor.py
  main.py
  mlflow.db
  model_pipeline.py
  requirements.txt
  streamlit.py
  streamlit.py
    load_model_pipeline
    lt, best_k_neighbors( "CLASSIFICATION PROBABILITIES:" )
    prob_df = pd.DataFrame([probabilities], columns=config.target_names)
    tf.dataframe(prob_df.style.format("{:.2%}"))

    except Exception as e:
      tf.error(f"An error occurred while compiling input features: {str(e)}")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER
warnings.warn(
  C:\Users\user\anaconda3\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
  C:\Users\user\anaconda3\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
  [I 2026-06-08 23:24:05.509] Trial 25 finished with value: 0.7237710947889718 and parameters: {'model_type': 'lightgbm', 'smote_k_neighbors': 4, 'lgb_n_estimators': 472, 'lgb_max_depth': 12, 'lgb_learning_rate': 0.1594705381811019, 'lgb_num_leaves': 136, 'lgb_subsample': 0.6499332829588534}. Best is trial 10 with value: 0.72448023755743298.
  [Early Stopping] Dibentikan otomatis pada Trial ke-25 karena tidak ada peningkatan setelah 15 trials.
```

File Edit Selection View Go Run Terminal Help

PROJECT: KAHIR MODEL DEPLOYMENT MLFLOW

streamlit.py

```
load_model_pipeline
tf.keras.models.load_model(
    probab_df = pd.DataFrame(probabilities, columns=config.target_names)
    tf.dataframe(probab_df.style.format("{:.2%}"))

except Exception as e:
    tf.error(f"An error occurred while compiling input features: {str(e)}")
```

Classification Report:

	precision	recall	f1-score	support
Poor	0.73	0.72	0.73	1446
Standard	0.75	0.76	0.76	2661
Good	0.64	0.62	0.63	893
accuracy			0.73	5000
macro avg	0.71	0.70	0.71	5000
weighted avg	0.73	0.73	0.73	5000

Ln 16, Col 16 Spaces: 4 UTF-8 LF Python 3.13.9 (base) Python 3.13 (64-bit) 23:25 08/06/2026

File Edit Selection View Go Run Terminal Help

PROJECT: KAHIR MODEL DEPLOYMENT MLFLOW

streamlit.py

```
load_model_pipeline
tf.keras.models.load_model(
    probab_df = pd.DataFrame(probabilities, columns=config.target_names)
    tf.dataframe(probab_df.style.format("{:.2%}"))
```

Figure 1

Confusion Matrix - Memakai Model Pipeline

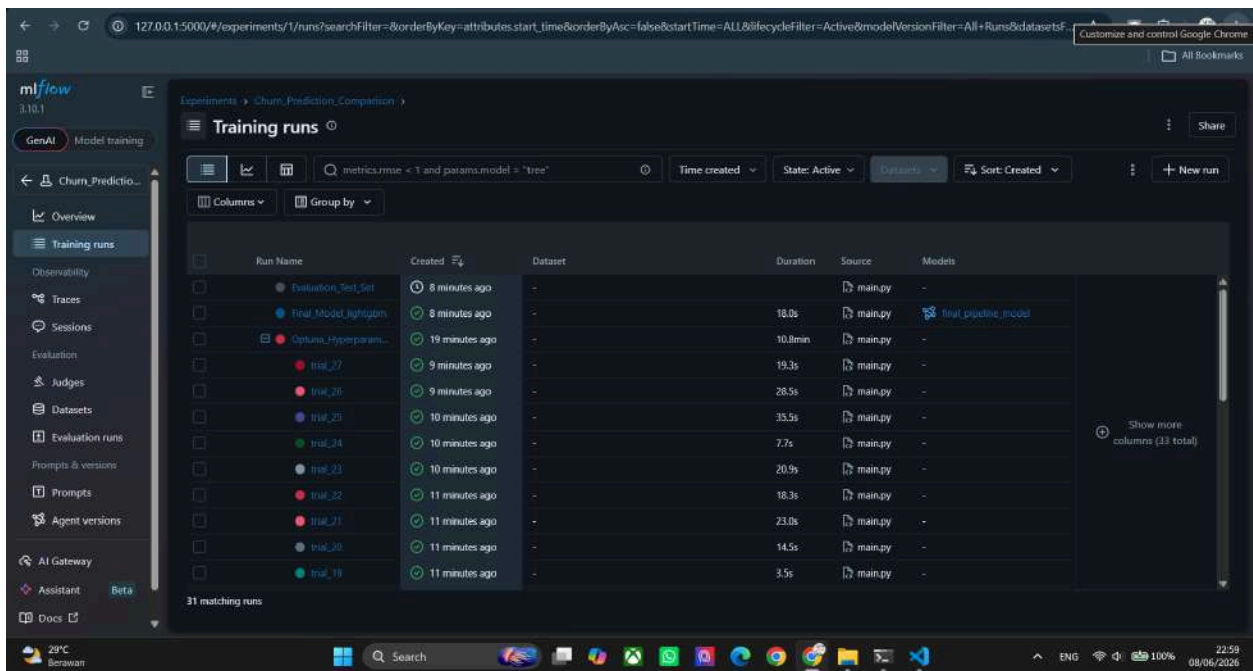
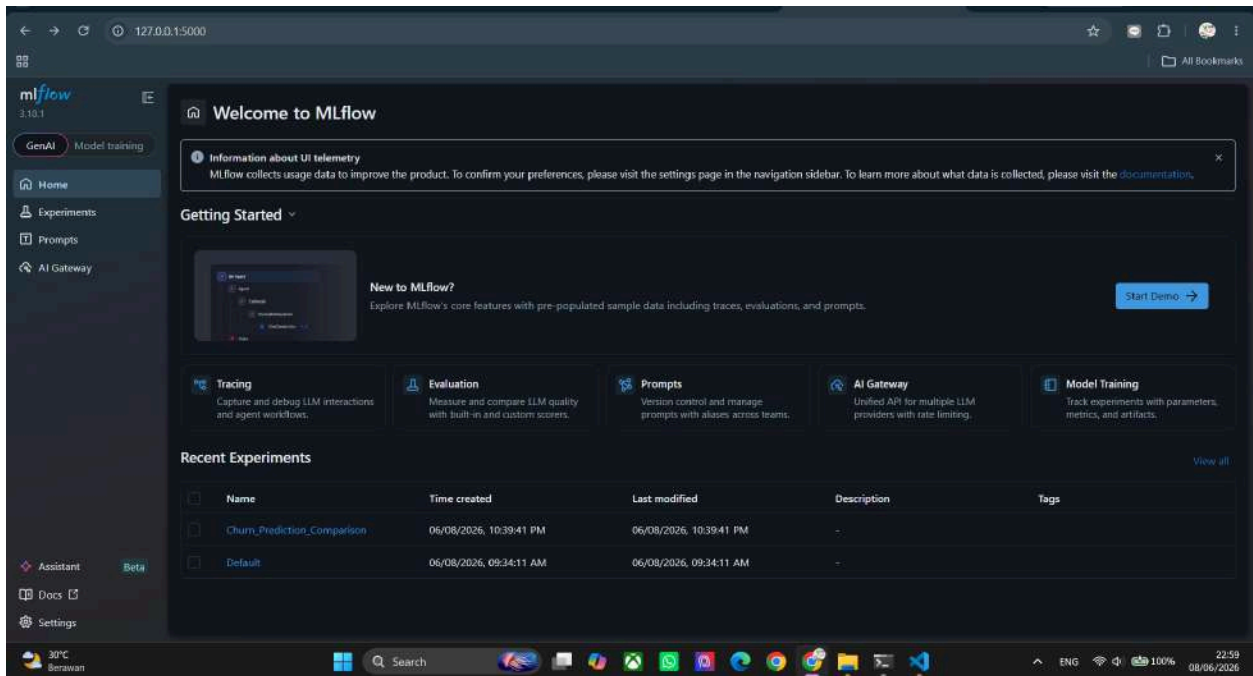
Kenyataan (Aktual) vs Prediksi Model

	Poor	Standard	Good
Poor	1047	350	49
Standard	363	2031	267
Good	21	315	557

Classification Report:

	precision	recall	f1-score	support
Poor	0.73	0.72	0.73	1446
Standard	0.75	0.76	0.76	2661
Good	0.64	0.62	0.63	893
accuracy			0.73	5000
macro avg	0.71	0.70	0.71	5000
weighted avg	0.73	0.73	0.73	5000

Ln 16, Col 16 Spaces: 4 UTF-8 LF Python 3.13.9 (base) Python 3.13 (64-bit) 23:24 08/06/2026



127.0.0.1:5000/#/experiments/1/runs/94179600ca8b4301abf5052de03661f8

mlflow

3.10.1

GenAI

Model training

Churn_Predictio...

Overview

Training runs

Observability

Traces

Sessions

Evaluation

Judges

Datasets

Evaluation runs

Prompts & versions

Prompts

Agent versions

AI Gateway

Assistant

Beta

Docs

Churn_Prediction_Comparison

Runs

Final_Model_lightgbm

Overview

Model metrics

System metrics

Traces

Artifacts

metrics (0)

No metrics recorded

Parameters (0)

No parameters recorded

Logged models (1)

Model attributes

Type	Step	Model name	Status	Created	Registered models
Output	0	final_pipeline_model	Ready	9 minutes ago	-

Experiment ID

Status

Run ID

Duration

Source

Registered prompts

Datasets

Tags

Registered models

1

Finished

94179600ca8b4301abf5052de03661f8

18.0s

main.py

-

None

Add tags

None

29°C

Borawan

Search

ENG

100%

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08/06/2026